Consensus-Based Memory Distillation: Preventing Agentic Drift in Autonomous LLM Systems

Cameron Warren
cwarre33@uncc.edu

Department of Computer Science
University of North Carolina at Charlotte

May 1, 2026

Abstract

Agentic systems that autonomously update their own memory suffer from *agentic drift*: a single hallucinated or biased interpretation can be permanently committed, compounding into cascading failures. We propose **Consensus-Based Memory Distillation**—applying Minimum Bayes Risk (MBR) decoding to memory formation by generating multi-model consensus before committing lessons to long-term storage. Our methodology creates a *memory confidence score* from candidate agreement, enabling risk-adjusted agent behavior when memory signal is unreliable. We formalize the approach mathematically and discuss implementation for autonomous trading and decision-making systems.

1 Introduction

Autonomous agentic systems must learn from experience. Standard practice uses single-pass LLM distillation: feed logs to a model, extract lessons, commit to knowledge base. This is vulnerable.

1.1 Agentic Drift

When a single LLM generation—potentially hallucinated or biased— is committed as *long-term memory*, subsequent decisions build on this corrupted foundation. Errors compound. The agent's "world model" drifts from reality.

Example: A trading agent observes an anomalous market event after a news spike. Single-pass distillation concludes: *"News spikes predict reversals."* This false pattern is committed to memory. Future positions sized according to confidence in this "lesson" underperform—but the agent cannot diagnose why: the corrupted memory is now part of its decision substrate.

1.2 Consensus-Based Approach

We apply **Minimum Bayes Risk (MBR) decoding** to memory formation:

1. Generate N independent distillation candidates from diverse models
2. Compute semantic similarity between candidates
3. Select the *consensus summary* with maximum expected utility
4. Commit to memory only if disagreement score is below threshold

This creates a principled *memory confidence score*:

$$\mathcal{C}(\text{memory}) = 1 - \frac{1}{N}\sum_{i=1}^{N} \text{dist}(s_i, s^*)$$

where s^* is the consensus and dist measures semantic divergence.

2 Related Work

MBR Decoding: Originally for machine translation, selecting output with highest expected BLEU against sample candidates. We generalize to semantic agreement for memory distillation.

LLM Memory Systems: Previous work on retrieval-augmented generation, memory-augmented networks, and continual learning. No prior work applies consensus mechanisms to prevent single-model hallucination in agent memory.

Multi-Model Ensembles: Existing approaches for classification and generation tasks. We specifically target *agentic* applications where memory permanence amplifies single-model errors.

3 Methodology

3.1 Consensus Memory Formation

Given log batch L, generate candidates:

$$\mathcal{S} = \{s_1, s_2, \ldots, s_N\}, \quad s_i \sim P_{\text{model}_i}(\cdot|L)$$

Using diverse providers (OpenAI GPT-4, Anthropic Claude, Meta Llama) to maximize independence.

3.2 Semantic Agreement

Embed each candidate:

$$e_i = \text{Embed}(s_i) \in \mathbb{R}^d$$

Compute similarity matrix:

$$M_{ij} = \cos(e_i, e_j)$$

The **consensus** is:

$$s^* = \arg\max_{s_i} \sum_{j=1}^{N} M_{ij}$$

3.3 Memory Confidence

$$\mathcal{C} = \frac{1}{N}\sum_{i=1}^{N} M_{i*}$$

where M_{i*} is similarity to consensus.

Threshold: Commit memory only if $\mathcal{C} \geq \tau$ (typically 0.75).

3.4 Risk-Adjusted Agent Behavior

Memory confidence modulates agent risk-taking:

- $\mathcal{C} \geq 0.9$: Full confidence—normal position sizing
- $0.75 \leq \mathcal{C} < 0.9$: Reduced confidence—smaller positions
- $\mathcal{C} < 0.75$: No commitment—human review required

4 Theoretical Properties

4.1 Anti-Hallucination

If a single model hallucinates pattern p not present in ground truth:

$$P(p \text{ in } s^*) \leq \frac{1}{N} \ll 1 \text{ for } N \geq 5$$

4.2 Bounded Drift

With confidence threshold τ, the maximum rate of false memory commitment is bounded by:

$$P(\text{false commit}) \leq P(\text{consensus hallucinates})$$

which decreases exponentially with N assuming model independence.

5 Implementation Considerations

5.1 Cost-Latency Tradeoff

N candidates implies $N\times$ inference cost. Mitigations:

- Use cheaper models (Llama 3.3 70B) for candidates
- Reserve frontier models (GPT-4, Claude) for final selection
- Batch process daily rather than real-time

5.2 Provider Independence

true independence requires diverse:

- Training data (different corpora)
- Architectures (transformer variants)
- Organizations (different alignment approaches)

6 Applications

6.1 Autonomous Trading

The original motivation: preventing corrupted market regime models from persisting in agent memory across trading cycles.

6.2 Research Agents

Scientific literature review agents where hallucinated paper summaries could propagate through citation chains.

6.3 Decision Support

Any agentic system learning from logged outcomes over extended operation periods.

7 Conclusion

Consensus-Based Memory Distillation extends MBR decoding to agentic memory formation, providing a principled defense against agentic drift. The confidence score $\mathcal{C}$ enables quantified risk-adjusted behavior when memory signal is uncertain.

Future work: empirical validation in production agentic systems, oracle studies comparing single-pass vs. consensus memory quality over extended operation windows.